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Roll No

EC-5005(3)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Industrial Electronics

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) How OP-AMP used in power supplies.
b) Describe Wien bridge oscillator, function generator.
2. The single phase full converter has RL load having $L = 6.5\text{mH}$, $R = 0.5\text{ ohm}$ and $E = 10\text{V}$ the input voltage is $V_s = 120\text{V}$ at (rms) 60Hz. Calculate:
 - i) Load current I_{LO} at $\omega t = \alpha = 60^\circ$
 - ii) The average thyristor current, I_A
 - iii) The average output current I_{dc}
3. a) Draw the block diagram of SMPS and explain each block in detail.
b) Explain the working of fly back and buck boost converters.
4. Explain Dynamic characteristics of SCR with the help of waveform and also explain T_{on} , T_{off} , $\frac{di}{dt}$, $\frac{dv}{dt}$ and t_c .
5. a) What are the factors to be consider during selection of PLC? What are the advantages of PLCs?
b) Explain briefly with a neat diagram, the architecture of a Programmable Logic Controller (PLC).

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6. Define the following terms related with thyristor.
 - i) Holding current
 - ii) Latching current
 - iii) Break over voltage
 - iv) di/dt and dv/dt
7.
 - a) Explain in brief the status and sequence of operations of PLC with the help of example.
 - b) What do you mean by ladder logic language explain in brief with the help of example.

OR

Write a short notes on the following:

- i) Fast recovery diode
 - ii) Triacs
 - iii) IGBT
8. Write short notes on any two of the following:
 - i) Power transistor
 - ii) Function generator using OP-Amp.
 - iii) Simple process control applications of PLC
